



Sony PlayStation 3: It's not all fun and games

- By William Jackson
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Studying gravitational radiation emitted by very massive black holes as they engulf stars takes a lot of computing power. Physics professor Gaurav Khanna at the University of Massachusetts Dartmouth has turned to Sony's popular PlayStation 3 gaming console to build his own supercomputer to help with the work.

Khanna and Glen Volkema of the physics department at UMass Dartmouth have built a cluster of eight of the consoles running Linux in parallel in Khanna's laboratory.

'My cluster is running code as fast as running 200 processors on a supercomputer,' Khanna said. Total cost of the homebuilt system would be under \$4,000 retail, the cost of running one large job on a supercomputer. However, most of the hardware for this project was donated.

Khanna is using the system to model the 'ripples in space and time' produced by black holes at the centers of galaxies in support of NASA's Laser Interferometer Space Antenna. LISA is being built in cooperation with the European Space Agency and will be launched in 2015 to detect gravitational waves, the death spirals of stars. Khanna is helping to determine what LISA will search for.

Khanna was attracted to PlayStation 3 by several unique features.

'One cool thing about it is that Sony made PS3 an open platform,' he said. 'That opened the possibility of using it for things beyond gaming.'

The second advantage is the Cell processor developed by Sony, IBM and Toshiba, which has 'great potential in raw computing power,' he said. Each PS3 processor is equal to about 25 processors in a traditional supercomputer.

Khanna began working on the project about a year ago.

'I was lucky that my wife was able to get one for me last Christmas,' when they were in short supply, he said. When he got it he was like 'well, like a kid on Christmas morning. 'I was so excited about it. As soon as I

got it I started taking it apart and putting Linux on it.'

This did not thrill his children, who wanted to play games on it, but 'now they have their own.'

And Khanna has eight of his own, most of them donated by Sony. He has received permission to spend department grant money on additional consoles and has six more on order. 'My goal is to have 16.'

Khanna's in-house system will not eliminate his need for time on more traditional supercomputers. Khanna last year was awarded a National Science Foundation grant for 30,000 hours of time on its TeraGrid computing infrastructure for his work on black holes and gravitational waves. However, using the time on a supercomputer means putting your job in queue and waiting until the necessary processors are available to run it. That can mean waiting days for a job that might take a few hours to run. 'It's very painful,' he said.

'The infrastructure hasn't kept up with demand,' he said. 'There is never enough computing time. So it's not that I won't be using my supercomputer time any more,' but he will not be as dependent on it.

Loading Linux onto the consoles and hooking them together to work in parallel was easier than he expected, Khanna said.

'I thought it was going to take some hacking, but it was fairly straightforward,' he said. 'Sony tells you how to do it.' The tricky part was rewriting his code to take advantage of the new Cell processor. 'I had to do a lot of learning. That took several months of work.'

Cell processors are good for supercomputing, and Los Alamos National Laboratory will be using them in a new peta-scale supercomputer it expects to have working next year. But inexpensive clusters of PS3s will not do away with the need for true supercomputers, Khanna said.

The fundamental limitation of the gaming console is its relatively limited memory, with only 512M of RAM per console, 'and it's not expandable,' he said.

'My problems are not heavy on memory usage,' he said. 'It's just heavy on raw calculation.' But if a job requires lots of memory, the PlayStation 3 probably won't do the trick.

About the Author

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